

Experimental Activity 1 (EA1): Planning, Localization, and Identification

Experimental activity EA1 involves planning, localization, and identification tasks on the actual setup. Following instructions reported below, you will have to:

1. Apply the ideal inputs necessary to follow an eight-shaper trajectory on the actual system, collecting wheels speed and robot configuration from the motion capture.
2. Use collected data to identify the robot parameters and calibration parameters, then simulate localization algorithms (offline).
3. Plan a trajectory that brings the robot from an initial configuration q_i to a final configuration q_f using cartesian polynomials. Run EKF and calibration during the trajectory execution.

You can deliver ONLY optional material in a single zip file. All the other exercises are propaedeutic to future activities. For this reason, it is important that you conclude them. If you encounter any issue, or results are not as you expected, please, discuss them with the professor or assistants.

Consider the following nominal parameters:

- Wheels radius: $r = 0.03 [m]$
- Wheels distance: $d = 0.165 [m]$
- Maximum wheels speed module: $\omega_M = 10 [rad/sec]$
- Sampling time $T_s = 0.04$

PART 1

1.1 Generate an eight-shaped trajectory. Use the following equations:

$$x(s) = R \sin(2\omega_{trj} s)$$

$$y(s) = R \sin(\omega_{trj} s)$$

where $s \in [0, 1]$ is the space variable, and $R = 0.4$, $\omega_{trj} = 2\pi i$. Compute also the derivatives analytically.

1.2 Use as a time law a trapezoidal time law. Consider as initial values $T_a = 1 [sec]$ and $T_c = 30 [sec]$.

1.3 Simulate the trajectory and control the evolution of x, y, θ, ω_L and ω_R as a function of time (use Simulink). Verify that wheels velocities modules are lower then ω_M .

- 1.4 Apply the inputs to the actual TurtleBot. **To prevent that the robot falls from the desk, keep the robot on its support platform, so that wheels run freely.**
- 1.5 Compare planned and actual wheels speed inputs.
- 1.6 Add the motion capture block to the Simulink file (ask for instructions to the laboratory assistants).
- 1.7 Prepare the Simulink file to save all variables necessary for identification and calibration.
- 1.8 Run the experiment with the following time law configurations:
 - $T_a = 1 \text{ [sec]}$ and $T_c = 30 \text{ [sec]}$
 - $T_a = 1 \text{ [sec]}$ and $T_c = 18 \text{ [sec]}$
 - $T_a = 1 \text{ [sec]}$ and $T_c = 45 \text{ [sec]}$

PART 2

- 2.1 Use data collected with the configuration $T_a = 1 \text{ [sec]}$ and $T_c = 30 \text{ [sec]}$ to identify the robot parameters and calibrate the robot and the motion capture.
- 2.2 Test the localization strategies (integration and EKF) by simulating collected data. In the EKF consider also possible measurement loss. Simulate all the time law configurations and evaluate eventual differences.

PART 3

- 3.1 Plan a trajectory that brings the robot from the initial configuration $q_i = \left[-0.5, -0.5, \frac{\pi}{2}\right]$ to the final configuration $q_f = \left[0.5, 0.5, \frac{\pi}{2}\right]$ using cartesian polynomials (in forward motion). Choose the time law such that the modules of wheels speeds are lower then ω_M .
- 3.2 Test the trajectory in simulation.
- 3.3 Apply the inputs to the actual TurtleBot and compare them with the planned one. **To prevent that the robot falls from the desk, keep the robot on its support platform, so that wheels run freely.**
- 3.4 Apply the input to the actual system, collecting data with the motion capture. The file must:
 - Calibrate data coming from the motion capture (remove offsets)
 - Run localization algorithm online (both integrations and EKF)
 - Save measures, commanded inputs, and state estimate
- 3.5 Compare collected results, eventually considering also different velocities.

Optional material to deliver

- Run 1.3-1.5, save results and write a script that plots the planned trajectory and compares planned and measured inputs. Name the script EX_opt_1.
- Run 2.2, save results and write a script that compares localization strategies. Consider all the three time law configurations. Name the script EX_opt_2.
- Run 3.4, save results, and write a script that compares:
 - o Planned and actual trajectory
 - o The trajectory estimated by the EKF considering a $p_{loss} = 1$, $p_{loss} = 0.99$ and $p_{loss} = 0$.

Name the script EX_opt_3.